P1

1. Suppose the odds is r, and if you win, you will get rb, otherwise, you lose your principal with bet b.

We let $p=0.85$, $q=0.6$ (market), so that:

$$q \cdot rb = (1-q) \cdot b \Rightarrow r = 2/3.$$

Thus, the utility for victor is:

$$\mathbb{E}U = p(d+rb) + (1-p)(d-b).$$

$$= d + b(rp - 1 + p)$$

while $rp - 1 + p = \frac{2}{3} \times 0.85 - 1 + 0.85 > 0$

$\mathbb{E}U$ is an increasing function over b.

So that Victor is going to maximize $\mathbb{E}U$ by letting

$$b^* = \max b = d.$$

And it ignores the risk of losing all principle, which does not make sense

2.

While $U(y) = \ln(y)$, we have:

$$\mathbb{E}U = p\log(d+rb) + (1-p)\log(d-b)$$

Now we maximize $\mathbb{E}U$:

$$\frac{\partial \mathbb{E}U}{\partial b} = \frac{pr}{d+rb} - \frac{1-p}{d-b} = 0$$

$$\Rightarrow b^* = d \cdot \frac{pr-(1-p)}{r}, \quad \frac{b^*}{d} = \frac{p-q}{1-q} = \frac{0.85-0.6}{1-0.85} = 0.625$$

which means:

Victor will bet 62.5% of his wealth

3. No. Without enough insider information, I don't trust the market's probability estimation. My utility is definitely loss-averse, so betting without an information is just pointless gambling. At most, I'd spend hundreds for fun.

P2

We have the cross entropy:

$$L(y, f) = -[y \log f + (1-y)\log(1-f)]$$

firstly, re write:

$$\mathbb{E}_{x,y}[L(y, f(x))] = \mathbb{E}_x[\mathbb{E}_{y|x}[L(y, f(x))]]$$

Denote, $p = P\{y=1 | x=x\}$, $q = f(x) \in [0,1]$, and

$$R(q) = \mathbb{E}_{y|x}[L(y, q)]$$

$$= p\cdot(-\log q) + (1-p)(-\log(1-q))$$

$$= -p\log q - (1-p)\log(1-q)$$

$$\Rightarrow R'(q) = -\frac{p}{q} + \frac{1-p}{1-q} = 0, \text{ we have: } q = p.$$

While $R''(q) = \frac{p}{q^2} + \frac{1-p}{(1-q)^2} > 0$, $R(q=p)$ is the unique maximum.

Such that:

$$f^*(x) = P(y=1 | x=x).$$

P3.

1. SGD uses noisy gradient estimates, with a constant step size $\rho_t = \rho$. we have:

$$\sum \rho_t^2 = \infty.$$

So that the iterates do not settle at the optimum, suppose k^*

Instead, they keep fluctuating within a neighborhood of it.

which is.

$$[k^* \pm \rho \nabla f]$$

2. While our goal is to minimize:

$$R(\theta) = \mathbb{E}[\mathcal{L}(Y, f(x, \theta))].$$

With only n samples, we actually minimize the empirical risk:

$$\hat{R}(\theta) = \frac{1}{n} \sum^n \mathcal{L}(y_i, f(x_i, \theta))$$

And the empirical minimizer $\hat{\theta}$ differs from the populations minimizer θ^* by a statistical error. which can not be removed by better optimizations

3. Roughly:

total error = statistical error + optimizations error.

So as long as the optimizations error by ρ, is no longer than the order of the statistical error, the total error is dominated by statistical error.

4. By choosing the constant ρ. we may gain other improvement:

1). Faster Iterations

2) Finite training budget.

3) So on ---